Favour O. Igwezeke

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Education

University of Nigeria, Nsukka — Nsukka, Nigeria
Doctor of Pharmacy (PharmD) — **Expected Feb. 2027**
CGPA: *4.22*/5.0 (Yr 1–5) · 4.59/5.0 (Yr 2–5)
Thesis: "Deep Learning–Based Behavioral and Anxiolytic Effects of the Methanol Stem Bark Extract of *Indigofera bracteolata* (DC.) in Wistar Rats"
Advisor: Assoc. Prof. Collins A. Onyeto, Dept. of Pharmacology & Toxicology, UNN

Research Interests

Biologically grounded representation learning; mechanistic and context-aware biomedical AI; uncertainty-aware reasoning and distribution shift in clinical and computational biology models; drug response prediction; perturbation modeling and transferability of biological representations.

Research Experience

SPARK (MICCAI-Endorsed Academy for African Medical Imaging Translation) — Remote
ML Research Fellow — Medical Imaging AI — Feb. 2026 – Sep. 2026

- **MORPHA** — Tested whether morphological priors improve cross-acquisition transfer in malaria microscopy; on Ugandan field data, reduced the transfer drop by 30.8% and halved calibration error, with a content-free control isolating morphology. *Presented at the* **MIRASOL Workshop @ MICCAI 2026**; *forthcoming in Springer LNCS.* [Paper]

Cancer Research with AI (CaresAI) — Remote
Medical AI Researcher — Jan. 2026 – Present
Group Lead: Dr. Mary Adewunmi

- **Mechanism or Memorisation?** — Introduced **cross-endpoint relational support (R_X)**, a training-relative diagnostic separating entity-level familiarity from relationship-level support in DTI prediction; showed that high-R_X interactions achieve consistently higher matched AP and lower adjusted loss across three representation families on 80,094 BindingDB interactions. *Accepted at the* **ICBINB-BIO Workshop @ NeurIPS 2026**. [Paper]
- **OmicsTrace** — Developed a leakage-controlled multi-omics framework that freezes pathway-anchored candidates before held-out RNA–protein confirmation in tumour/adjacent proteogenomics, extending to RA synovial transcriptomics and PubMed knowledge mining; showed that curated lung cancer pathways **confirm no better than size- and abundance-matched random gene programs**. *In preparation for the* **GeNeDis 2026 Hackathon**.
- **CURE** — Developed an uncertainty-aware Graph-RAG system for evidence-grounded multihop biomedical question answering; used **selective claim-level evidence verification**, backtracking, and abstention to study reliability under incomplete evidence. *Manuscript in preparation for* **Transactions on Machine Learning Research (TMLR)**.
- **CT-DEB'26** — Developed an ensemble ML framework integrating four domain-specific BERT embeddings to predict medication dosing errors across 29,000+ ClinicalTrials.gov protocols. *Published at* **CL4Health @ LREC 2026**. [Paper]

Dept. of Pharmacology & Toxicology, University of Nigeria, Nsukka — Nsukka, Nigeria
Undergraduate Researcher — Computational Toxicology & Neuropharmacology — Nov. 2024 – Present
Supervisor: Assoc. Prof. Collins A. Onyeto

- **CellStateGRN** *(ongoing)* — Formulating a **perturbation-anchored causal representation framework** to identify shared regulatory programs and sparse state-specific rewiring across compensating and proliferating hepatocyte states in liver repair; planned validation against synthetic ground truth and evaluation on interventions withheld from model fitting.
- **TrajRisk** — Developed a **multi-scale directed GNN learning treatment-aware transitions** across hepatocyte injury trajectories (36,067 cells); resolved temporally distinct recovery phases distinguishing genuine resolution from persistent compensation and transferred zero-shot across species and modalities to human liver-failure injury regions. *Manuscript complete; to be submitted to* **PLOS Computational Biology**.
- **Undergraduate Thesis:** Investigated anxiolytic effects of *Indigofera bracteolata* using DeepLabCut-based pose estimation (>94% accuracy); quantified dose-dependent effects (Cohen's d=0.8–1.3 at 600 mg/kg) across three behavioral paradigms; automated pipeline reduced analysis time 6×. *Paper under review at* **PLOS ONE**.
- Examined structure–toxicity relationships in 30 NSAIDs via unsupervised chemoinformatics; found non-hepatotoxic NSAIDs cluster within hepatotoxic chemical neighborhoods, implicating mechanistic over structural determinants of DILI. [Code]

African Society for Bioinformatics & Computational Biology / NIH ODSS — Remote

Bioinformatics Researcher Dec. 2025 – Aug. 2026

- **absuite** — Developed an R/Bioconductor package (10 functions) for antibody repertoire profiling; validated on 3.4M sequences across COVID-19 and influenza cohorts, reproducing published convergent clonotypes at 98.8% concordance. *Presented at* **ISMB 2026** (oral), **BioC 2026** (oral), and **EuroBioC 2026** (poster). Package in preparation for Bioconductor submission.

Dept. of Pharmaceutical & Medicinal Chemistry, University of Nigeria, Nsukka Nsukka, Nigeria
Undergraduate Researcher — Computational Drug Discovery Sept. 2025 – Present
Supervisor: Prof. Charles O. Nnadi

- **CASPID** — Designed a context-aware neural conditioning framework across 23,248 EGFR/BRAF inhibitor–cell line pairs; learns context-dependent structural feature importance from cellular transcriptomic state, revealing interpretable MET-mediated bypass resistance patterns. *Manuscript in preparation for BMC Bioinformatics.* [Code]
- **TransQSAR-pf** — Developed a biology-informed QSAR framework integrating *P. falciparum* transcriptomic signatures with 125 compounds; 71.2% of predictive performance attributed to conserved stress-response genes over chemical descriptors alone. *Published in MDPI Engineering Proceedings.* [Paper] [Code] [Poster]

HackBio Remote
Bioinformatics Research Intern — Next Generation Sequencing Sept.–Nov. 2025
Supervisor: Dr. Adewale Ogunleye, PhD

- Performed guilt-by-association co-expression analysis on 69 HCC samples, identifying stage-specific lncRNAs tracking nivolumab+ipilimumab response, enriched in IFN/T-cell pathways, and validating a 12-lncRNA panel for monitoring immunotherapy progression. [Paper] [Code]

Genomac Institute Inc. Remote
Cancer Genomics Research Intern Jul.–Oct. 2025

- Performed multi-omics analysis of 123 meningioma samples (TCGA, cBioPortal), integrating somatic mutation and CNA data to validate TMB and CNA as candidate biomarkers for WHO-grade stratification. *Poster presented at ISPOR-HEOR UNN 2025 and CoGSAYR 2026.* [Code] [Poster] [Poster]

Person-Centered HIV Research Team (PeCHIVReT), University of Nigeria, Nsukka Nsukka, Nigeria
Undergraduate Researcher — Clinical Epidemiology Sept. 2024 – Present
Supervisor: Dr. AbdulMuminu Isah, PharmD, PhD

- Extracted and synthesized safety data from 113 HIV vaccine trials (1991–2024) for a systematic review; contributed to statistical analysis, evidence mapping, and manuscript preparation. *Paper under review at BMC Infectious Diseases.*

HackBio Remote
Bioinformatics Intern — Cancer Genomics Sept.–Nov. 2024

- Analyzed TCGA-BRCA and GBM RNA-Seq cohorts (1,200+ samples); identified three biomarkers (*CST5*, *CGA*, *MYL9*) with RF classification; GO/KEGG enrichment revealed biosynthetic dysregulation in gliomas. [Portfolio]

Publications

Bold = author; † equal contribution (co-first author); * corresponding author.

Peer-Reviewed Publications & Conference Papers

[1] **Igwezeke, F. O.**, & Adewunmi, M. A. (2026). Mechanism or Memorisation? Stress-Testing Relational Generalisation in Drug–Target Interaction Prediction. *I Can't Believe It's Not Better: Failure Modes of AI in Biology* (***ICBINB-BIO Workshop @ NeurIPS 2026***). *Accepted poster.*

[2] **Igwezeke, F. O.**, Ugwuishiwu, C. H., Emesiani, J. U., Agada, S. I., Anozie, E. L., Kama, M. O., & Emegoakor, A. C. (2026). MORPHA: Morphology-Constrained Training and the Limits of Cross-Acquisition Transfer in Low-Resource Malaria Microscopy. ***MIRASOL Workshop @ MICCAI 2026***. To appear in the *Lecture Notes in Computer Science* series (Springer).

[3] Abubakar, J. I., Jarme, J., **Igwezeke, F.**, & Adewunmi, M. A. (2026). CaresAI at SMM4H-HeaRD 2026: Predicting TNM Staging. In *Proceedings of the 11th Social Media Mining for Health Research and Applications Workshop and Shared Tasks* (***SMM4H-HeaRD 2026 @ ACL 2026***).

[4] Hamnett, L., **Igwezeke, F.**, Abubakar, J. I., & Adewunmi, M. A. (2026). CaresAI at CT-DEB'26: Detecting Dosing Errors in Clinical Trials Using Domain-Specific Transformer Embeddings and Classification Models. In *Proceedings of the Third Workshop on Patient-Oriented Language Processing* (***CL4Health 2026 @ LREC 2026***).

[5] **Igwezeke, F. O.** & Nnadi, C. O. (2026). TransQSAR-pf: A Bio-Informed QSAR Framework Using Plasmodium falciparum Stress Signatures for Enhanced Antiplasmodial Activity Prediction. *Engineering Proceedings*, 124(1), 37.

Preprints & Manuscripts Under Review

[6] **Igwezeke, F. O.**, & Onyeto, C. A. (2026). TrajRisk: Treatment-Aware Multi-Scale Graph Neural Networks for Trajectory-Informed Risk Propagation in Single-Cell Transcriptomics. *Manuscript complete; to be submitted to PLOS Computational Biology.*

[7] **Igwezeke, F. O.**, Abdel-Salam, R., Idagu, E., & Adewunmi, M. (2026). CURE: Selective Evidence Verification for Multihop Biomedical Graph-RAG Question Answering. *In preparation for submission to Transactions on Machine Learning Research (TMLR).*

[8] Ogunleye, A. J.[*], El-Tony, Y.[*], Adegite, A. A.[*], **Igwezeke, F. O.**[†], Mahdi, S. M.[†], Mohamed, M. A.[†], Orimolade, T.[†], Osatoyinbo, O.[†] (2026). Exploring lncRNAs as Biomarkers for Monitoring Hepatocellular Carcinoma Immunotherapy Response. *Research Square*. Manuscript under review at *Gene*.

[9] Onyeto, C. A., **Igwezeke, F. O.**, Uwakwe, C. C., et al. (2026). Deep Learning-Based Behavioural Analysis of Anxiolytic-Like Activity of *Indigofera bracteolata* Methanol Stem Extract in Wistar Rats. *Under review*, *PLOS ONE*.

[10] Ayeni, V. A., Azumara, S. I., Agbo, C. E., **Igwezeke, F. O.**, et al. (2025). Safety Profile of HIV Vaccine Candidates Across Clinical Trial Phases: A Scoping Review. *Under review*, *BMC Infectious Diseases*.

Awards, Honours & Fellowships

Oct. 2026 **NeurIPS 2026 Financial Assistance Award**, Neural Information Processing Foundation

Jun. 2026 **ISMB 2026 Conference Fellowship**, International Society for Computational Biology (ISCB) — Competitive conference fellowship award ($1,500 USD)

Jun. 2026 **BioC 2026 Travel Scholarship**, Bioconductor Conference — $550 USD travel award + complimentary registration

Mar. 2026 **Okwuchukwu Fund Research Award**, UNN — NGN 120,000 competitive grant for undergraduate thesis research (1 of 3 recipients)

Dec. 2025 **ABI Travel Fellowship**, African Bioinformatics Institute — Full support, CoGSAYR Africa Summit, Lagos

Nov. 2025 🏆 **Gabriel Ezenri Award for Excellence in Research 2.0 — 2nd Place** (NGN 110,000), ISPOR-HEOR UNN

Feb. 2025 🏆 **Gabriel Ezenri Award for Excellence in Research — 1st Place** (NGN 100,000), ISPOR-HEOR UNN

Feb. 2025 **1st Runner-up**, ISPOR-HEOR UNN Clinical Skills Evaluation Competition

Nov. 2024 **Research Fellow**, Toufik's World Medical Association Research Fellowship in Neurology

Nov. 2024 **Udacity × Bertelsmann Next Generation Tech Booster Scholarship** — Top 500 worldwide ($1,200 USD)

Sept. 2024 **HackBio Precision Medicine Premium Scholarship** — Selected 5 of 200+ applicants

Oct. 2023 **YandyTech × DataCamp Donates Data Science Scholarship** ($399 USD)

Software

absuite — R/Bioconductor package for antibody repertoire profiling and clonotype analysis

10 analytical modules; validated on 3.4M BCR sequences across COVID-19 and influenza cohorts; in preparation for Bioconductor submission.

OmicsTrace — Python/R framework for held-out multi-omics confirmation of candidate genes

Snakemake workflow; frozen development/confirmation splits, prespecified baselines and matched random-program controls; applied to CPTAC lung proteogenomics and RA synovial transcriptomics; public release in preparation.

Selected Presentations

Oral Presentations

[1] absuite: An R/Bioconductor Package for Antibody Repertoire Profiling and Clonotype Analysis. **BioC 2026** (Bioconductor Conference), Seattle, WA, Aug. 2026. *Short talk.* [Slides]

[2] absuite: A Package in R for Antibody Repertoire Profiling and Clonotype Analysis. **ISMB 2026** (International Conference on Intelligent Systems for Molecular Biology), Jul. 2026. [Slides]

[3] Anxiolytic Effects of *Indigofera bracteolata* Using Deep Learning-Based Automated Behavioral Analysis. *1st Int. Conference on Pharmaceutical, Herbal & Cosmetic Products*, UNN, Feb. 2026. [Slides]

[4] Ensemble ML with Uncertainty Quantification for Pharmacogenetic-Guided Warfarin Dosing Across Diverse Populations. *CoGSAYR Africa Summit 2026*, Lagos, Jan. 2026.

[5] Ensemble ML with Uncertainty Quantification for Pharmacogenetic-Guided Warfarin Dosing Across Diverse Populations. *ISPOR-HEOR UNN Conference*, Nov. 2025. 🏆 **2nd Place, Best Oral Presentation.** [Web App]

[6] Unsupervised ML Framework for IDH Status Prediction in Low-Grade Gliomas. *5th Annual National Conference on Bioscience & Biotechnology*, UNN, Jul. 2025.

[7] Exploratory Transcriptomic Analysis Reveals Preferential Downregulation of RNA Polymerase II in Recurrent Glioblastoma. *ISPOR-HEOR UNN Conference*, Feb. 2025. 🏆 **1st Place, Best Oral Presentation.** [Paper]

Poster Presentations

[8] MORPHA: Morphology-Constrained Training and the Limits of Cross-Acquisition Transfer in Low-Resource Malaria Microscopy. *MIRASOL Workshop @ MICCAI 2026*, Strasbourg, France, Sep. 2026. [Poster]

[9] absuite: An R/Bioconductor Package for Antibody Repertoire Profiling and Clonotype Analysis. *EuroBioC2026*, Turku, Finland, Jun. 2026.

[10] Comparative Analysis of TMB and CNA in WHO-Graded Meningiomas. *CoGSAYR Africa Summit 2026*, Lagos, Jan. 2026. [Poster]

[11] TransQSAR-pf: A Bio-Informed QSAR Framework Using *P. falciparum* Stress Signatures. *6th Int. Electronic Conference on Applied Sciences (ASEC 2025)*, p. 24, Dec. 2025. [Poster]

[12] Transcriptomic Subtype Stratification of Low-Grade Gliomas by IDH Status. *NABIS 2025*, Ogbomoso, Oct. 2025. *NABIS 2025 Proceedings*, pp. 24–25.

[13] Safety Profile of Candidate HIV Vaccines Across Clinical Trial Phases. *13th IAS Conference on HIV Science*, Kigali, Rwanda, Jul. 2025. *IAS 2025 Book of Abstracts*, p. 328.

Service & Teaching

Academic Service

Reviewer, Africa in AI Workshop at NeurIPS 2026
Reviewer, Machine Learning for Health (ML4H) Symposium 2026
Scientific Program Committee Member, Machine Learning in Computational Biology (MLCB) 2026

Teaching Experience

Peer Academic Tutor, University of Nigeria, Nsukka 2024–2025
Tutored 37 students in genetics, chemistry, and statistics; co-created 30+ study guides and question banks used by over half of the Pharmacy Class of 2026.

Technical Skills

Programming	Python (PyTorch, PyG, scikit-learn, XGBoost, Hugging Face, RDKit, Scanpy), R (Bioconductor, limma, edgeR, DESeq2, Monocle 3), SQL
ML Methods	GNNs; protein/chemical language models (ESM-2, ChemBERTa); Graph-RAG; uncertainty and selective prediction (MC dropout, calibration); causal representation learning; pose estimation (DeepLabCut)
Comp. Biology	sc/snRNA-seq, trajectory inference, cross-species mapping; bulk RNA-seq and CPTAC proteogenomics; DTI and QSAR; antibody repertoires (AIRR); cancer genomics (TCGA)
Evaluation	Leakage-controlled splits, frozen analysis plans, matched random controls, ablations, bootstrap CIs
Tools	Git, Docker, Snakemake, HPC, AWS, Streamlit

Professional Development

In progress	**Data Structures and Algorithms Specialization**, UC San Diego (Coursera) — graph algorithms, dynamic programming, string algorithms
In progress	**Foundations of Probability and Statistics Specialization**, University of Colorado Boulder (Coursera) — probability, statistical inference, Markov chains, Monte Carlo
Aug. 2026	**Mathematics for Machine Learning Specialization**, Imperial College London (Coursera) — linear algebra, multivariate calculus, PCA
Jul. 2025	**Cohere Labs ML Summer School** — Transformers, diffusion models, self-supervised learning; faculty from Google DeepMind, Meta AI (FAIR), Adobe, INRIA, MIT. [Certificate]
Jun. 2025	**Compute Ontario Summer School** — HPC, bioinformatics pipelines (RNA-Seq), cloud computing, Docker/Git. [Certificate]